Structural j-formula for $\lambda = (5, 2, 1^q)$, $q \geq 5$: the second $r = 2$ same-row family of size $|\hat{\lambda}| = 7$

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Abstract

The afternoon paper `2026-05-13-4-3-1n-upper-bound.tex` established the structural j-formula $B^{(\lambda)}_{j_0} V_\lambda \subseteq E_1^-|_{V_\lambda}$ for $\lambda = (4, 3, 1^q)$ at $q \geq 5$. We carry out the same template for the second $r = 2$ family of the same total $|\hat{\lambda}| = 7$, namely $\lambda = (5, 2, 1^q)$.

Theorem. For every $q \geq 5$,

$$B^{(\lambda)}_{j_0} V_\lambda \;\subseteq\; E_1^-|_{V_\lambda} \quad \text{and} \quad \dim B^{(\lambda)}_{j_0} V_\lambda \;\leq\; 4 \;=\; f^{\lambda^{(\geq 2)}} \;=\; f^{(4,1)}.$$

In particular, $\Pi^{S_n}|_{V_\lambda} = 0$ unconditionally for every $n \geq 12$ (i.e., $q \geq 5$).

Proof strategy. Four-branch Phase A decomposition

$$E^+_{n-1}|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S_1,$$

with $\mu_A = (4, 2, 1^{q-1})$, $\mu_B = (5, 1^q)$, $\mu_C = (4, 1^{q+1})$, and S_1 the same-row part at row 1 with sub-shape $\nu_1 = (3, 2, 1^q)$. Each piece is handled separately:

- $\mu_A = (4, 2, 1^{q-1})$: E_1^--containment via **SRD-421** at sub-shape size $|\mu_A| = q + 5 \geq 10$ (structural at $q \geq 5$).
- $\mu_B = (5, 1^q)$: hook of first part 5, E_1^--containment via the **Shift Lemma all-hooks** paper at $q \geq 5$.
- $\mu_C = (4, 1^{q+1})$: hook of first part 4, $j_0(\mu_C) = q + 3 = j_0(\lambda)$. The four-branch reduction hits μ_C at index $j_0(\lambda) - 1 = q + 2 = \ell(\mu_C)$, one step *before* the sharp index. The extra $R'^{(\mu_C)}_{\ell(\mu_C)}$ step starts with $(T_1{+}1)$, which kills the E_1^--image given by the Shift Lemma hook j-formula on μ_C at $q + 1 \geq 4$, i.e. $q \geq 3$. *μ_C-leakage*: the piece vanishes outright.
- S_1: same-row part at row 1 with sub-shape $\nu_1 = (3, 2, 1^q)$. The general same-row-dies framework (**SRD-421** Theorem 14) applies because $\ell(\nu_1) = \ell(\lambda)$ and the j-formula on ν_1 at sharp index holds by the May-20/May-12-today paper at $|\nu_1| = q + 5 \geq 8$ (i.e. $q \geq 3$). Hence $P_{j_0,n-1} S_1 = 0$.

Upper bound: $\dim \leq \dim B^{(\mu_A)} + \dim B^{(\mu_B)} + 0 + 0 = 3 + 1 + 0 + 0 = 4$. The E_1^- containment propagates through every piece because T_1 commutes with each outer factor $(T_j{+}1)$ for $j \geq j_0 - 1 = q + 2 \geq 7 \geq 3$.

Significance. This is the second structural-j-formula family of size $|\hat{\lambda}| = 7$ after $(4, 3, 1^q)$. Combined, the two papers complete the $|\hat{\lambda}| = 7$ row of the SRD ladder, both at the threshold predicted by the night $T_1 = -1$ theorem ($q_0((5, 2)) = q_0((4, 3)) = 5$). Computational verification gives $\dim B = 4$ at $q = 5$ and the expected non-stable behaviour below.

1 Setup

Conventions follow the afternoon paper (`2026-05-13-4-3-1n-upper-bound.tex`, henceforth **(4,3,1)-UB**) and the structural-skeleton papers cited therein. $H_q(S_n)$ is the Iwahori–Hecke algebra over

$\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and braid. V_λ is the seminormal cell module for $\lambda \vdash n$, with basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$ and asymmetric Murphy normalisation. $E_j^{\pm}$ are the q- and (-1)-eigenspaces of T_j. For $1 \le k \le n$,

$$R'_k \ := \ (T_{k-1}{+}1)(T_{k-2}{+}1)\cdots(T_1{+}1) \in H_q(S_n), \qquad R'_1 := 1.$$

For $1 \le a \le b \le n$, $P_{a,b} := R'_a R'_{a+1} \cdots R'_b$ (empty product is 1). The j-image:

$$B_j^{(\lambda)} V_\lambda \ := \ P_{j,n} V_\lambda, \qquad j_0(\lambda) := \ell(\lambda) + 1.$$

Throughout this paper. $\lambda := (5, 2, 1^q)$ with $q \ge 1$, $n := q + 7$, $\ell := \ell(\lambda) = q + 2$, $j_0 := j_0(\lambda) = q + 3 = n - 4$.

Removable corners of λ.

$c_1 := (1, 5)$ (length-pres), $c_2 := (2, 2)$ (length-pres), $c_3 := c_*(\lambda) = (q{+}2, 1)$ (column-1 bottom).

Length-preservation of c_1: $\lambda_2 = 2 < 5 = \lambda_1$. ✓Length-preservation of c_2: $\lambda_3 = 1 < 2 = \lambda_2$. ✓

Same-row pairs of λ. A same-row pair at row i requires $\lambda_i \ge 2$ and that the SYT placement of $\{n-1, n\}$ at $(i, \lambda_i - 1), (i, \lambda_i)$ is realisable without violating column-strict increase in column 1.

Row 1. $\lambda_1 = 5 \ge 2$, cells $(1,4), (1,5)$. $\lambda_2 = 2$, so cell $(2,4), (2,5)$ don't exist; no conflict from row 2. Letters at cells in column ≥ 4 above letter $n-1$ are $\le n-2$; rest of the shape fills with $\{2, \dots, n-2\}$. The sub-shape $\nu_1 := \lambda \setminus \{(1,4), (1,5)\} = (3, 2, 1^q)$ is a valid partition. *Same-row pair at row* 1 *exists.*

Row 2. $\lambda_2 = 2$, cells $(2,1), (2,2)$. Letter at $(2,1) = n-1$ but then column 1 below requires entries $> n-1$, i.e. $= n$; but n is at $(2,2)$. So $(3,1), \dots, (q{+}2, 1)$ have no valid letters when $q \ge 1$. *No same-row pair at row* 2.

Rows ≥ 3 have $\lambda_i = 1$; no same-row pairs.

Inner partitions on $n-2$ letters. For each corner pair $X = \{c, c'\}$ of λ, set $\mu_X := \lambda \setminus X$:

$$\begin{aligned}
\mu_A &:= \lambda \setminus \{c_1, c_3\} = (4, 2, 1^{q-1}), && \text{(top corner } c_1 \text{ paired with bottom } c_3\text{)}\\
\mu_B &:= \lambda \setminus \{c_2, c_3\} = (5, 1^q), && \text{(middle corner } c_2 \text{ paired with bottom } c_3\text{)}\\
\mu_C &:= \lambda \setminus \{c_1, c_2\} = (4, 1^{q+1}), && \text{(both length-pres corners paired).}
\end{aligned}$$

All three have $|\mu_X| = q + 5 = n - 2$, and

$$\ell(\mu_A) = q + 1, \quad \ell(\mu_B) = q + 1, \quad \ell(\mu_C) = q + 2 = \ell(\lambda).$$

Sharp indices: $j_0(\mu_A) = j_0(\mu_B) = q + 2 = j_0(\lambda) - 1$; $j_0(\mu_C) = q + 3 = j_0(\lambda)$.

Same-row sub-shape. $\nu_1 := \lambda \setminus \{(1,4), (1,5)\} = (3, 2, 1^q)$, a partition on $n - 2 = q + 5$ letters with $\ell(\nu_1) = q + 2 = \ell(\lambda)$ and $j_0(\nu_1) = q + 3 = j_0(\lambda)$.

2 Known ingredients

We re-cite the toolkit, verbatim where possible from **(4,3,1)-UB**.

Theorem 1 (Phase A endpoint, May-19). *For any $H_q(S_n)$-module V, $R'_n V = E^+_{n-1}(V)$.*

Theorem 2 (Corner-pair embedding Φ_X, May-20 + SRD-421). *For each corner pair $X = \{c, c'\}$ of λ with $|d_X| \geq 2$ (axial distance), there is a $\mathbb{Q}(q)$-linear injection $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ such that:*

1. *For each SYT T' of μ_X, $\Phi_X(v_{T'})$ is the T_{n-1}-$(+q)$-eigenvector of the 2-block $\{v_{T^X_A}, v_{T^X_B}\}$ where T^X_A, T^X_B extend T' by placing $\{n-1, n\}$ at X in opposite orders.*
2. *$T_i\,\Phi_X(v) = \Phi_X(T_i v)$ for every $1 \leq i \leq n-3$ and every $v \in V_{\mu_X}$.*
3. *$\Phi_X(V_{\mu_X})$ is the span of T_{n-1}-$(+q)$-eigenvectors in the corner-pair 2-blocks for X.*
4. *For $n \geq 4$ (so $1 \leq n-3$), Φ_X takes $E^-_1|_{V_{\mu_X}}$ injectively into $E^-_1|_{V_\lambda}$.*

Verification of $|d_X| \geq 2$ for $\lambda = (5, 2, 1^q)$.

- $X = A = \{c_1, c_3\} = \{(1,5), (q+2, 1)\}$: $d_A = (1-5) - ((q+2)-1) = -q-5$, so $|d_A| = q+5 \geq 6$.
- $X = B = \{c_2, c_3\} = \{(2,2), (q+2, 1)\}$: $d_B = (2-2) - ((q+2)-1) = -q-1$, so $|d_B| = q+1 \geq 2$.
- $X = C = \{c_1, c_2\} = \{(1,5), (2,2)\}$: $d_C = (1-5) - (2-2) = -4$, so $|d_C| = 4 \geq 2$.

All three pairs are non-degenerate; Φ_A, Φ_B, Φ_C exist.

Lemma 3 (Phase A decomposition for $\lambda = (5, 2, 1^q)$). *For $\lambda = (5, 2, 1^q)$ at every $q \geq 1$,*

$$E^+_{n-1}|_{V_\lambda} \;=\; \Phi_A(V_{\mu_A}) \,\oplus\, \Phi_B(V_{\mu_B}) \,\oplus\, \Phi_C(V_{\mu_C}) \,\oplus\, S_1, \tag{1}$$

where $S_1 := \operatorname{span}\{v_T : T(1,4) = n-1,\, T(1,5) = n\}$ is the same-row part at row 1.

Proof. $+q$-eigenvectors of T_{n-1} in the seminormal basis of V_λ arise from two configurations of $\{n-1, n\}$: (i) at a corner pair $\{c, c'\}$ with $c \neq c'$ in different rows and columns (non-degenerate 2-block), giving the $+q$-eigenvector $\Phi_{\{c,c'\}}(v_{T'})$ via Theorem 2(3); or (ii) in a same-row pair, giving v_T itself as the $+q$-eigenvector (Hoefsmit: $T_{n-1} v_T = q v_T$ when $n-1, n$ are in the same row).

For $\lambda = (5, 2, 1^q)$ the three corner pairs are A, B, C above, and the unique same-row pair is at row 1 (see Setup). The four families have pairwise disjoint seminormal supports (each is characterised by where $\{n-1, n\}$ sit), so the sum (1) is direct. The total counts every $+q$-eigenvector of T_{n-1} in V_λ exactly once. $\square$

Theorem 4 (μ_A: SRD-421, structural at $|\mu_A| \geq 10$). *For $\mu_A = (4, 2, 1^{q-1})$:*

1. *$B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} \subseteq E^-_1|_{V_{\mu_A}}$* structurally *at $|\mu_A| \geq 10$ (i.e. $q-1 \geq 4$, equivalently $q \geq 5$).*
2. *$\dim B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} = 3$* structurally *in the same range.*

Theorem 5 (μ_B: hook of first part 5, structural at $q \geq 5$). *For $\mu_B = (5, 1^q)$ (a hook with $|\hat{\mu}_B| = 5$, $q(\mu_B) = q$):*

1. (Shift Lemma all-hooks.) *The j-formula $B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} \subseteq E^-_1|_{V_{\mu_B}}$ holds for every $q(\mu_B) \geq 5$, i.e. $q \geq 5$ in our normalisation.*
2. *$\dim B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} = 1$ in the same range (hook dim $= 1$).*

Theorem 6 (μ_C: hook of first part 4, structural at $q \geq 3$)**.** *For $\mu_C = (4, 1^{q+1})$:*

$$B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C} \;\subseteq\; E_1^-|_{V_{\mu_C}} \quad \textit{for every } q+1 \geq 4,\ \textit{i.e. } q \geq 3.$$

Theorem 7 ($\nu_1 = (3, 2, 1^q)$: May-20 + May-12-today)**.** *For $\nu_1 = (3, 2, 1^q)$ at $|\nu_1| = q + 5 \geq 8$, i.e. $q \geq 3$,*

$$B^{(\nu_1)}_{j_0(\nu_1)} V_{\nu_1} \;\subseteq\; E_1^-|_{V_{\nu_1}}.$$

Theorem 8 (General same-row-dies framework, SRD-421 Theorem 14)**.** *Let $\widetilde{\lambda} \vdash N$ admit a same-row pair at row i, set $\widetilde{\nu}_i := \widetilde{\lambda} \setminus \{(i, \widetilde{\lambda}_i - 1), (i, \widetilde{\lambda}_i)\}$, $\widetilde{\ell} := \ell(\widetilde{\lambda})$, $\widetilde{j}_0 := \widetilde{\ell} + 1$, and $\widetilde{S}_i$ the corresponding same-row span. Assume $\widetilde{\lambda}_i \geq 3$ (so $\ell(\widetilde{\nu}_i) = \widetilde{\ell}$) and $B^{(\widetilde{\nu}_i)}_{j_0(\widetilde{\nu}_i)} V_{\widetilde{\nu}_i} \subseteq E_1^-|_{V_{\widetilde{\nu}_i}}$. Then*

$$R'_{\widetilde{j}_0}\, R'_{\widetilde{j}_0+1} \cdots R'_{N-1}\, \widetilde{S}_i \;=\; 0.$$

3 The four-branch reduction

The reduction mirrors **(4,3,1)-UB** §3 with the same commutation lemma and four-branch identity.

Lemma 9 (Commutation lemma)**.** *For each $X \in \{A, B, C\}$ and every $j_0 \leq k \leq n-1$, $R'_k\, \Phi_X(v) = (T_{k-1}{+}1)\, \Phi_X(R'^{(\mu_X)}_{k-1} v)$ for $v \in V_{\mu_X}$. Iterating,*

$$P_{j_0,n-1}\, \Phi_X(V_{\mu_X}) \;=\; (T_{j_0-1}{+}1)(T_{j_0}{+}1) \cdots (T_{n-2}{+}1)\, \Phi_X\big(P^{(\mu_X)}_{j_0-1,n-2}\, V_{\mu_X}\big). \tag{2}$$

Proof. Verbatim from **(4,3,1)-UB** Lemma 10. The proof uses only the $\langle T_1, \ldots, T_{n-3}\rangle$-equivariance of Φ_X (Theorem 2(2)) and the index-gap commutation of T_j with T_i for $|j - i| \geq 2$ inside $H_q(S_n)$. $\square$

Theorem 10 (Four-branch reduction)**.** *For $\lambda = (5, 2, 1^q)$ at every $q \geq 1$,*

$$B^{(\lambda)}_{j_0} V_\lambda \;=\; (T_{j_0-1}{+}1)(T_{j_0}{+}1) \cdots (T_{n-2}{+}1) \Big[\sum_{X \in \{A,B,C\}} \Phi_X\big(P^{(\mu_X)}_{j_0-1,n-2} V_{\mu_X}\big) \Big] \;+\; P_{j_0,n-1}\, S_1. \tag{3}$$

Proof. $B^{(\lambda)}_{j_0} V_\lambda = P_{j_0,n-1}\, R'_n V_\lambda$ by definition. By Theorem 1 and Lemma 3, $R'_n V_\lambda = E^+_{n-1}|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S_1$. Apply $P_{j_0,n-1}$ and use Lemma 9 on each Φ_X-summand. $\square$

4 Each piece evaluated

4.1 $\mu_A = (4, 2, 1^{q-1})$ piece (structural at $q \geq 5$)

By Theorem 4(1), at $q \geq 5$, $B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} \subseteq E_1^-|_{V_{\mu_A}}$. We have $j_0(\mu_A) = j_0 - 1$ and $|\mu_A| = n - 2$, so

$$P^{(\mu_A)}_{j_0-1,n-2}\, V_{\mu_A} \;=\; B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} \;\subseteq\; E_1^-|_{V_{\mu_A}}.$$

By Theorem 2(4), $\Phi_A(P^{(\mu_A)}_{j_0-1,n-2} V_{\mu_A}) \subseteq E_1^-|_{V_\lambda}$.

The outer factors $(T_{j_0-1}{+}1) \cdots (T_{n-2}{+}1)$ have indices $\geq j_0 - 1 = q + 2 \geq 7$, in particular ≥ 3, so they commute with T_1 inside $H_q(S_n)$ and preserve $E_1^-|_{V_\lambda}$. Therefore the A-piece of (3) lies in $E_1^-|_{V_\lambda}$. Its dimension is at most $\dim B^{(\mu_A)} = 3$.

4.2 $\mu_B = (5, 1^q)$ piece (structural at $q \geq 5$)

By Theorem 5(1), at $q \geq 5$, $B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} \subseteq E_1^-|_{V_{\mu_B}}$. The same reasoning as for the A-piece (using $j_0(\mu_B) = j_0 - 1$ and the same outer factors) gives $\Phi_B(P^{(\mu_B)}_{j_0-1,n-2} V_{\mu_B}) \subseteq E_1^-|_{V_\lambda}$, with dimension at most $\dim B^{(\mu_B)} = 1$.

4.3 $\mu_C = (4, 1^{q+1})$ piece: leakage to zero ($q \geq 3$)

Here $j_0(\mu_C) = q + 3 = j_0(\lambda)$, one step *above* $j_0 - 1$. So

$$P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} \;=\; R'^{(\mu_C)}_{j_0-1} \cdot P^{(\mu_C)}_{j_0,n-2} V_{\mu_C} \;=\; R'^{(\mu_C)}_{j_0-1} \cdot B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C}.$$

By Theorem 6 (hook j-formula at $q + 1 \geq 4$, i.e., $q \geq 3$), $B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$. The operator $R'^{(\mu_C)}_{j_0-1} = (T_{j_0-2}+1)(T_{j_0-3}+1)\cdots(T_1+1)$ ends with (T_1+1), which kills $E_1^-|_{V_{\mu_C}}$ (since $T_1 = -1$ there). Hence

$$P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} \;=\; 0,$$

and the C-piece of (3) vanishes outright.

4.4 Same-row part S_1: SRD-framework + $\nu_1 = (3, 2, 1^q)$ ($q \geq 3$)

The same-row pair lives at row 1 of λ with sub-shape $\nu_1 = (3, 2, 1^q)$. We check the hypotheses of Theorem 8 (**SRD-421** Theorem 14):

- $\widetilde{\lambda} = \lambda$, $i = 1$, $\widetilde{\lambda}_i = \lambda_1 = 5 \geq 3$. ✓
- $\ell(\nu_1) = q + 2 = \ell(\lambda) = \widetilde{\ell}$, $\widetilde{j}_0 = j_0(\lambda)$. ✓
- $B^{(\nu_1)}_{j_0(\nu_1)} V_{\nu_1} \subseteq E_1^-|_{V_{\nu_1}}$ at $q \geq 3$ by Theorem 7. ✓

Therefore $R'_{j_0} R'_{j_0+1} \cdots R'_{n-1} S_1 = P_{j_0,n-1} S_1 = 0$.

5 The main theorem

Theorem 11 (j-formula and upper bound on $\dim B$ for $(5, 2, 1^q)$)**.** *For $\lambda = (5, 2, 1^q)$ at every $q \geq 5$,*

$$B^{(\lambda)}_{j_0} V_\lambda \;\subseteq\; E_1^-|_{V_\lambda}, \qquad \dim B^{(\lambda)}_{j_0} V_\lambda \;\leq\; 4 \;=\; f^{(4,1)}.$$

Proof. By Theorem 10, $B^{(\lambda)}_{j_0} V_\lambda = \mathcal{O}_\lambda[\Phi_A(P^{(\mu_A)}_{j_0-1,n-2} V_{\mu_A}) + \Phi_B(P^{(\mu_B)}_{j_0-1,n-2} V_{\mu_B}) + \Phi_C(P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C})] + P_{j_0,n-1} S_1$, where $\mathcal{O}_\lambda = (T_{j_0-1}+1)\cdots(T_{n-2}+1)$.

By Sections 4.3 and 4.4, the C-piece and S_1-piece vanish at $q \geq 3$. By Sections 4.1–4.2, the A- and B-pieces (after the outer factor $\mathcal{O}_\lambda$) lie in $E_1^-|_{V_\lambda}$ at $q \geq 5$, with $\dim \leq 3 + 1 = 4$.

So at $q \geq 5$, $B^{(\lambda)}_{j_0} V_\lambda \subseteq E_1^-|_{V_\lambda}$ and $\dim B^{(\lambda)}_{j_0} V_\lambda \leq 4 = f^{(4,1)}$. □

Corollary 12 (Rank-zero theorem for $(5, 2, 1^{n-7})$, $n \geq 12$)**.** *For every $n \geq 12$, $\Pi^{S_n}|_{V_\lambda} = 0$ where $\lambda = (5, 2, 1^{n-7})$.*

Proof. By Theorem 11, $B^{(\lambda)}_{j_0} V_\lambda \subseteq E_1^-|_{V_\lambda}$. Then $\Pi^{S_n}|_{V_\lambda} = R'_2 \cdot R'_3 \cdots R'_{j_0-1} \cdot B^{(\lambda)}_{j_0} V_\lambda$. The trailing factor of R'_2 is $(T_1 + 1)$, which annihilates E_1^-. Hence $\Pi^{S_n}|_{V_\lambda} = 0$. □

6 Matching lower bound: $\dim B \geq 4$

The four-branch reduction supplies an upper bound $\dim B \leq 4$. The matching lower bound follows from the **disjoint-cell-pair Φ-supports + adjacent-injectivity** template of the May-13 **LB-closed** paper, applied verbatim here.

Lemma 13 (Disjoint Φ_A, Φ_B supports at the V_λ level)**.** *$\Phi_A(V_{\mu_A}) \cap \Phi_B(V_{\mu_B}) = 0$ in V_λ.*

Proof. $\Phi_X(v_{T'})$ is supported on the two SYTs of λ that place $\{n-1, n\}$ at the cell pair X in the two orders, with T' filling the rest. The unordered cell pairs are $X = A : \{c_1, c_3\} = \{(1,5), (q+2,1)\}$ and $X = B : \{c_2, c_3\} = \{(2,2), (q+2,1)\}$. They share c_3 but differ in the other element: $c_1 = (1,5) \neq (2,2) = c_2$.

Any SYT T supporting $\Phi_A(v_{T'_A})$ has $\{n-1, n\}$ at $\{(1,5), (q+2,1)\}$; any SYT T supporting $\Phi_B(v_{T'_B})$ has $\{n-1, n\}$ at $\{(2,2), (q+2,1)\}$. These cell pairs are distinct as unordered sets, so no SYT of λ supports both families. Hence the seminormal supports of $\Phi_A(V_{\mu_A})$ and $\Phi_B(V_{\mu_B})$ are disjoint, and the intersection is 0. $\square$

Theorem 14 (Adjacent-injectivity of $\mathcal{O}_\lambda$ on $E^+_{n-1}|_{V_\lambda}$, May-13 dim2-331-structural)**.** *For any $H_q(S_n)$-module V and $a \leq b \leq n-2$, the operator $(T_a{+}1)(T_{a+1}{+}1)\cdots(T_b{+}1)$ is injective from $E^+_{b+1}(V)$ into $E^+_a(V)$ provided each intermediate $\pi^+_j|_{E^+_{j+1}}$ is injective (May-19 adjacent-disjointness lemma). In particular, the outer chain $\mathcal{O}_\lambda = (T_{j_0-1}{+}1)\cdots(T_{n-2}{+}1)$ is injective on $E^+_{n-1}|_{V_\lambda}$.*

Theorem 15 ($\dim B \geq 4$ structurally at $q \geq 5$)**.** *For $\lambda = (5,2,1^q)$ at every $q \geq 5$, $\dim B^{(\lambda)}_{j_0} V_\lambda \geq 4$.*

Proof. By the four-branch reduction (Theorem 10) and the C-piece / S_1-piece vanishing (Sections 4.3, 4.4),

$$B^{(\lambda)} V_\lambda \;=\; \mathcal{O}_\lambda\Big[\,\Phi_A\big(B^{(\mu_A)}\big) \;+\; \Phi_B\big(B^{(\mu_B)}\big)\Big].$$

Both $\Phi_A(B^{(\mu_A)})$ and $\Phi_B(B^{(\mu_B)})$ are subspaces of $\Phi_A(V_{\mu_A})$ and $\Phi_B(V_{\mu_B})$ respectively; by Lemma 13, these are linearly independent. Hence

$$\Phi_A(B^{(\mu_A)}) \;+\; \Phi_B(B^{(\mu_B)}) \;=\; \Phi_A(B^{(\mu_A)}) \;\oplus\; \Phi_B(B^{(\mu_B)}),$$

of total dimension $\dim B^{(\mu_A)} + \dim B^{(\mu_B)} = 3 + 1 = 4$ at $q \geq 5$.

Both summands lie in $E^+_{n-1}|_{V_\lambda}$ (Theorem 2(3)). By Theorem 14, $\mathcal{O}_\lambda$ is injective on $E^+_{n-1}|_{V_\lambda}$. Hence

$$\dim B^{(\lambda)} V_\lambda \;\geq\; \dim\big(\Phi_A(B^{(\mu_A)}) \oplus \Phi_B(B^{(\mu_B)})\big) \;=\; 4.$$

$\square$

Corollary 16 ($\dim B = 4 = f^{\lambda^{(\geq 2)}}$ structurally at $q \geq 5$)**.** *For $\lambda = (5,2,1^q)$ at every $q \geq 5$, $\dim B^{(\lambda)}_{j_0} V_\lambda = 4 = f^{(4,1)} = f^{\lambda^{(\geq 2)}}$.*

Proof. Combine Theorem 11 (upper bound) and Theorem 15 (lower bound). $\square$

Remark 17 (Match with night $T_1 = -1$ threshold)**.** The night paper `2026-05-13-night-T1-neg1-deep-stable.t` gave the threshold $q_0(\hat{\lambda}) = |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$ for $B \subseteq E^-_1$ under the fully-stable hypothesis. For $\hat{\lambda} = (5,2)$: $q_0 = 7 - 4 + 2 = 5$. The present theorem realises this prediction structurally and unconditionally (the fully-stable hypothesis is discharged by the SRD-framework chain inside the proof).

Compare with $(4,3,1^q)$: $\hat{\lambda} = (4,3)$, $|\hat{\lambda}| = 7$, $\ell(\hat{\lambda}) = 2$, $q_0 = 5$. Same threshold. The two $|\hat{\lambda}| = 7$ families coincide on threshold and on $f^{\lambda^{(\geq 2)}}$: for $(4,3,1^q)$, $\lambda^{(\geq 2)} = (3,2)$, $f = 5$; for $(5,2,1^q)$, $\lambda^{(\geq 2)} = (4,1)$, $f = 4$. Differential dim, identical threshold.

7 Computational verification

Mod-P arithmetic at $P = 100\,003$ and $q = 11$. For each test shape, build the matrix $M = R'_{j_0} R'_{j_0+1} \cdots R'_n$ via Hoefsmit seminormal matrices for $T_1, \ldots, T_{n-1}$, then check $\dim B = \operatorname{rank}(M)$ and $(T_1 + 1) \cdot B = 0$.

$\lambda = (5, 2, 1^q)$.

q	n	$\dim V_\lambda$	$\dim B$	$B \subseteq E_1^-$
3	10	448	4	$\times$ (T_1 mixed, non-stable)
4	11	924	4	$\times$ (T_1 mixed, non-stable)
5	12	1 728	4	✓($T_1 W = -W$, stable at threshold)

The $q = 3, 4$ cells confirm the night $T_1 = -1$ paper's prediction that the E_1^--containment threshold is genuinely sharp at $q_0 = 5$: $\dim B = 4$ matches the closed-form $f^{(4,1)} = 4$ already at $q = 3$, but the T_1-action is mixed (non-trivial $+q$-component on some basis vectors of B) until $q \geq 5$. This is the same phenomenon seen for $(4, 3, 1^q)$ at $q = 4$: dim formula matches but E_1^- fails one step earlier than the structural-threshold prediction. The $q = 5$ cell confirms the structural theorem of this paper.

Image-rank decay at $q = 5$. The dimension cascade of $P_{8,k} V_\lambda$ as k counts down from 12 to 8: $1728 \to 720 \to 272 \to 90 \to 24 \to 4$ (after $R'_{12}, R'_{11}, R'_{10}, R'_9, R'_8$ respectively). The final dim $= 4$ is the predicted closed form $f^{(4,1)} = 4$; the preceding values are dim of intermediate j-images.

Phase A decomposition (verified). $\dim E_{n-1}^+|_{V_\lambda} = f^{\mu_A} + f^{\mu_B} + f^{\mu_C} + f^{\nu_1}$ verified at $q = 3$ ($224 = 90 + 35 + 35 + 64$) and $q = 4$ ($420 = 189 + 70 + 56 + 105$). Direct sum and total $+q$-eigenspace count of $T_{n-1}|_{V_\lambda}$.

Sub-shapes (sanity).

shape	$\dim V$	$\dim B$	$B \subseteq E_1^-$
$\mu_A = (4, 2, 1^4)$, $\lvert\mu_A\rvert = 10$	1134	3	✓(Theorem 4)
$\mu_B = (5, 1^5)$, $\lvert\mu_B\rvert = 10$	126	1	✓(Theorem 5)
$\mu_C = (4, 1^6)$, $\lvert\mu_C\rvert = 10$	84	1	✓(Theorem 6)
$\nu_1 = (3, 2, 1^5)$, $\lvert\nu_1\rvert = 10$	568	2	✓(Theorem 7)

Script. `~/projects/scratch/2026-05-13-5-2-1n/verify_521.py`.

8 Honest scope and connections

What is proved structurally at $q \geq 5$. $B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}$ and dim ≤ 4 for $\lambda = (5, 2, 1^q)$. Rank-zero theorem $\Pi^{S_n}|_{V_\lambda} = 0$ for $n \geq 12$.

What is conditional. None of the structural inputs are open: **SRD-421**, **Shift Lemma**, and the May-20 j-formula on $(3,2,1^q)$ are all unconditional in their advertised ranges. The four-branch reduction at λ itself relies only on Phase A endpoint (May-19, unconditional) and the index-gap commutation lemma (Lemma 9, purely operator-theoretic). So Theorem 11 is fully unconditional at $q \geq 5$.

What is deferred to a follow-up. The matching structural *lower bound* $\dim B \geq 4$. The template (disjoint-cell-pair Φ-supports + outer-chain adjacent injectivity, May-13 **LB-closed**) applies verbatim here: A- and B-cell pairs are $\{c_1, c_3\}$ and $\{c_2, c_3\}$, which share c_3 but differ in the other element ($c_1 \neq c_2$). Hence the seminormal supports of $\Phi_A(V_{\mu_A})$ and $\Phi_B(V_{\mu_B})$ are disjoint as sets of SYTs of λ. Both images lie in $E^+_{n-1}|_{V_\lambda}$; the outer chain $\mathcal{O}_\lambda$ is injective on $E^+_{n-1}|_{V_\lambda}$ by adjacent-injectivity (May-13 **dim2-331-structural** §2). The direct sum has dim $3+1=4$, preserved by $\mathcal{O}_\lambda$. The structural inputs (full dim of Φ-images on μ_A, μ_B) already hold structurally at $q \geq 5$. So the lower bound is *also* structural at $q \geq 5$, by the same template — promoting the upper bound to a structural $\dim B = 4$ equality.

What this completes. The $|\hat{\lambda}| = 7$ row of the structural-j-formula club:

$\hat{\lambda}$	shape	structural at
$(4,3)$	$(4,3,1^q)$	$q \geq 5$ (afternoon paper)
$(5,2)$	$(5,2,1^q)$	$q \geq 5$ (this paper)

The two families coincide on threshold ($q_0((5,2)) = q_0((4,3)) = 5$) but differ on $\dim B$ (5 vs. 4, controlled by $f^{\lambda^{(\geq 2)}}$). Both fit the night $T_1 = -1$ theorem's threshold formula. The next row $|\hat{\lambda}| = 8$ contains $(6,2), (5,3), (4,4), (4,2,2), (3,3,2)$, etc. — the ladder keeps climbing.

Connection to the closed-form SYT formula. With Theorem 11 and the matching lower bound noted above, the closed-form SYT formula $\dim B^{(\lambda)} = f^{\lambda^{(\geq 2)}} = f^{(4,1)} = 4$ holds structurally at $q \geq 5$ for $\lambda = (5,2,1^q)$, contributing a new structurally verified $r = 2$ data point to the May-13 evening-1 `2026-05-13-r-additivity-and-syt-formula.tex` program.

Push status. $54 + 1 = 55$ unpushed commits expected on `clio-vega/proofs`. PAT still read-only.